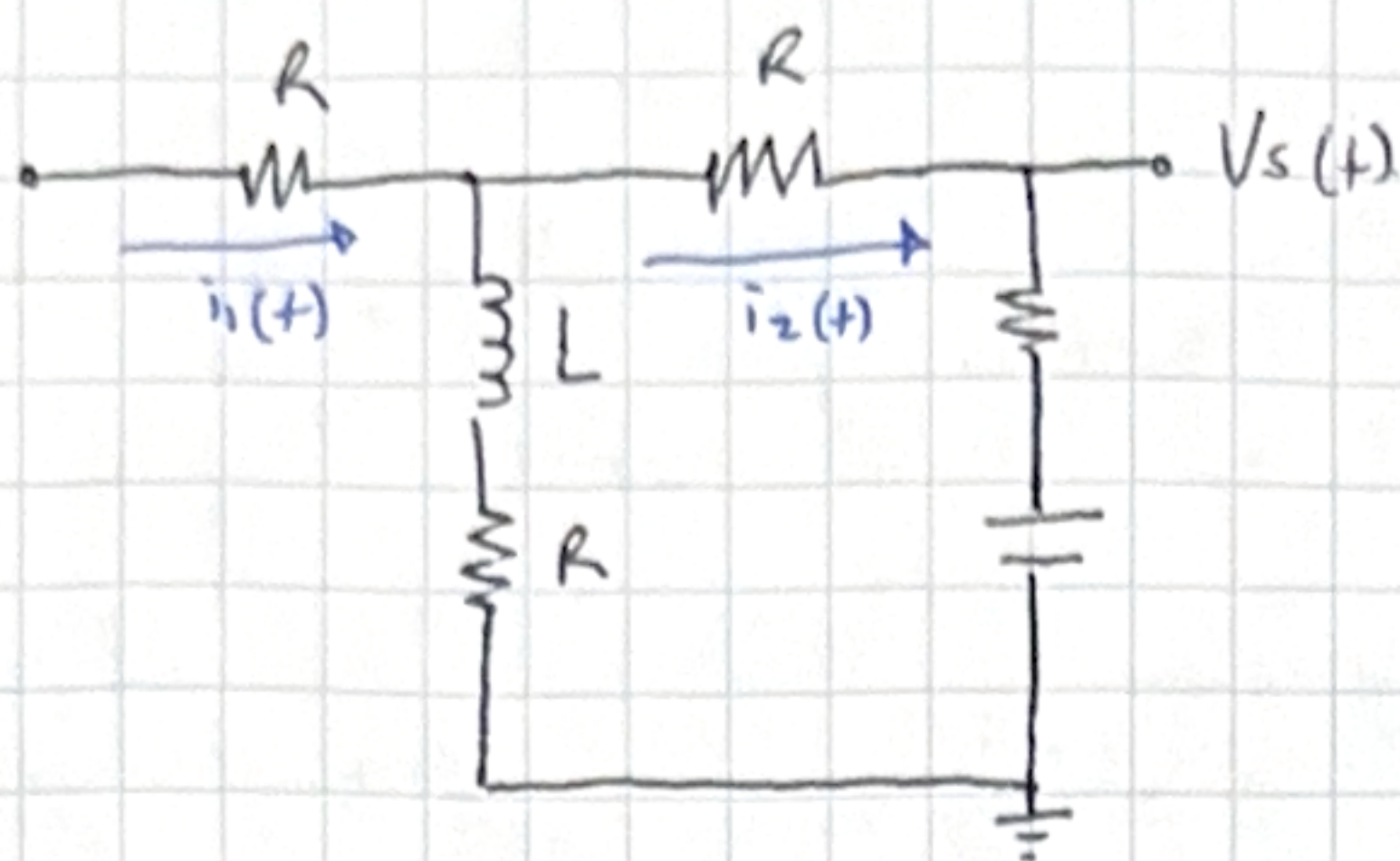


5.1. Practica: Diseño de controlador para un sistema de segundo orden



Ecuaciones principales

$$V_e(t) = R i_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)]$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + R[i_1(t) - i_2(t)] = R i_2(t) + R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$V_s(t) = R i_2(t) + \frac{1}{C} \int i_2(t) dt$$

Transformada de Laplace

$$V_e(s) = R I_1(s) + L s [I_1(s) - I_2(s)] + R[I_1(s) - I_2(s)]$$

$$L s [I_1(s) - I_2(s)] + R[I_1(s) - I_2(s)] = R I_2(s) + R I_2(s) + \frac{I_2(s)}{C s}$$

$$V_s(s) = R I_2(s) + \frac{I_2(s)}{C s}$$

Procedimiento algebraico

orden: polinomio + división +

$$V_e(s) = (L s + 2R) I_1(s) - (L s + R) I_2(s)$$

$$I_1(s) = \frac{1}{L s + R} \left(L s + \frac{1}{C s} + 3R \right) I_2(s) = \frac{C L s^2 + 3 C R s + 1}{C L s^2 + C R s} I_2(s)$$

$$V_s(s) = \left(R + \frac{1}{C s} \right) I_2(s) = \frac{C R s + 1}{C s} I_2(s)$$

Función de transferencia

$$\frac{V_s(s)}{V_e(s)} = \frac{(C R s + 1 / C s) I_2(s)}{\left[(L s + 2R) \frac{C L s^2 + 3 C R s + 1}{C L s^2 + C R s} - (L s + R) \right] I_2(s)}$$

$$= \frac{\frac{C R s + 1}{C s} I_2(s)}{\frac{3 C L R s^2 + (5 C R^2 + L) s + 2 R}{C L s^2 + C R s} I_2(s)}$$

$$= \frac{(C R s + 1)(L s + R)}{3 C L R s^2 + (5 C R^2 + L) s + 2 R}$$

Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} s \left(1 - \frac{C L R s^2 + (C R^2 + L) s + R}{3 C L R s^2 + (5 C R^2 + L) s + 2 R} \right) = \frac{R}{2 R}$$

$$= \frac{1}{2} V$$

Estabilidad en lazo abierto

$$R = 4.7 \text{ k}\Omega$$

$$C = 330 \mu\text{F}$$

$$L = 10 \text{ mH}$$

$$a = 3CLR = 3(330 \times 10^{-6})(10 \times 10^{-3})(4.7 \times 10^3) = 0.04653$$

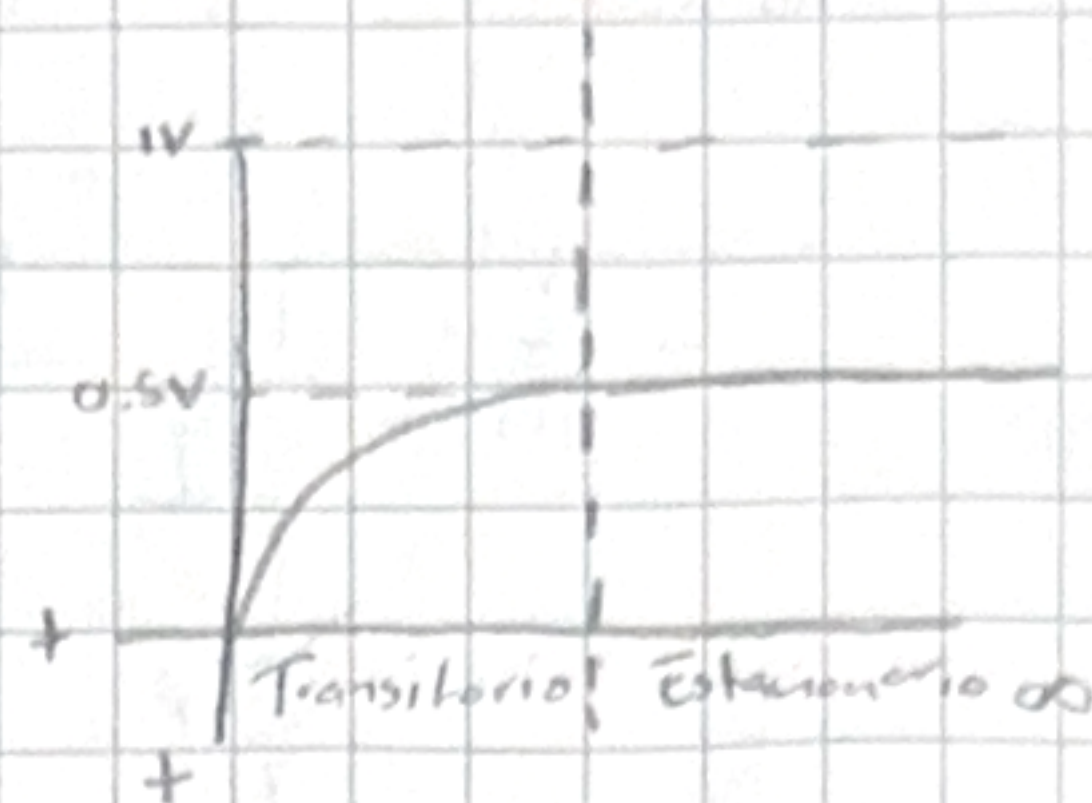
$$b = 5CR^2 + L = 5(330 \times 10^{-6})(4.7 \times 10^3)^2 + 10 \times 10^{-3} = 36448.51$$

$$c = 2R = 2(4.7 \times 10^3) = 9400$$

$$\lambda_{1,2} = \frac{-36448.51 \pm \sqrt{(36448)^2 - 4(0.04653)(9400)}}{2(0.04653)}$$

$$\lambda_1 = -0.257897$$

$$\lambda_2 = -783333.8061$$



Subamortiguado

Modelo de ecuaciones integra-diferenciales

El modelo se expresa despijando las variables dependientes, es decir, las corrientes

$$V_c(t) = 2Ri_1(t) + L \frac{d[i_1(t) - i_2(t)]}{dt} - Ri_2(t)$$

$$i_1(t) = \left[V_c(t) + Ri_2(t) - L \frac{d[i_1(t) - i_2(t)]}{dt} \right] \frac{1}{2R}$$

$$L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) = Ri_2(t) + Ri_2(t) + Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$

$$i_2(t) = \left[L \frac{d[i_1(t) - i_2(t)]}{dt} + Ri_1(t) - \frac{1}{C} \int i_2(t) dt \right] \frac{1}{3R}$$

$$V_c(t) = Ri_2(t) + \frac{1}{C} \int i_2(t) dt$$